Number Systems

The big idea

Numbers grew as people needed them: natural numbers for counting, zero, negatives (integers), fractions (rationals) and finally numbers like √2 that can't be written as fractions (irrationals). Together, rationals and irrationals make the **real numbers**, which fill the whole number line.

THE FAMILY OF NUMBERS

Natural $N = \{1, 2, 3, \ldots\} \subset$ Whole $W = \{0, 1, 2, \ldots\} \subset$ Integers $Z = \{\ldots, -1, 0, 1, \ldots\} \subset$ Rational Q $(p/q, q \neq 0) \subset$ Real R.
Irrational numbers ($\sqrt{2}$, $\sqrt{3}$, π) are real but not rational.

Between any two rational numbers there are **infinitely many** rational numbers. For example, between 1 and 2: 1.1, 1.01, 1.001, … or use $(a + b)/2$ repeatedly.

Decimal expansions

DECIMALS TELL YOU THE TYPE

Rational: decimal expansion is **terminating** ($1/4 = 0.25$) or **non-terminating recurring** ($1/3 = 0.333\ldots$, $1/7 = 0.142857142857\ldots$).
Irrational: **non-terminating non-recurring** ($\sqrt{2} = 1.41421356\ldots$, $\pi = 3.14159\ldots$).

WRITE 0.4747… AS P/Q

Let $x = 0.4747\ldots$ Then $100x = 47.4747\ldots$
Subtract: $99x = 47 \Rightarrow$ **$x = 47/99$**.

Real numbers on the number line

√2 can be located exactly: draw a unit square on the number line from 0 to 1; its diagonal has length √2 (Pythagoras). With a compass centred at 0, mark the diagonal's length on the line. Repeating this (a right triangle with legs √2 and 1 gives √3) builds the **square-root spiral**.

Operations and rationalising

Rational + irrational = irrational; non-zero rational × irrational = irrational. But two irrationals can give a rational: √2 × √2 = 2.

IDENTITIES WITH ROOTS (A, B > 0)

$$\sqrt{(ab)} = \sqrt{a}\sqrt{b},\ \sqrt{(a/b)} = \sqrt{a}/\sqrt{b}$$

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$$

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

RATIONALISE 1/(√7 − √2)

Multiply top and bottom by (√7 + √2): (√7 + √2)/(7 − 2) = **(√7 + √2)/5**.

Laws of exponents for real numbers

FOR A > 0 AND RATIONAL P, Q

$$a^p \cdot a^q = a^{(p+q)},\ (a^p)^q = a^{(pq)},\ a^p/a^q = a^{(p-q)},\ a^p b^p = (ab)^p,$$

$$a^0 = 1,\ a^{(1/n)} = \sqrt[n]{a},\ a^{(m/n)} = (\sqrt[n]{a})^m$$

SIMPLIFY 64^(1/2) AND 32^(2/5)

64^(1/2) = √64 = **8**. 32^(2/5) = $(\sqrt[5]{32})^2 = 2^2 = $ **4**.

Quick recap

- Real numbers = rationals + irrationals.
- Terminating or recurring decimal ⇒ rational; non-terminating non-recurring ⇒ irrational.
- Rationalise using $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$.

Practice questions

1. Which of these is irrational?
   (a) 0.75 (b) 22/7 (c) √5 (d) 0.333…
2. The decimal expansion of 1/8 is
   (a) terminating (b) non-terminating recurring (c) non-terminating non-recurring (d) not possible
3. 0.666… expressed as p/q is
   (a) 6/10 (b) 2/3 (c) 66/100 (d) 3/2
4. (√3 + √2)(√3 – √2) equals
   (a) 1 (b) 5 (c) √6 (d) 0
5. The value of 125^(1/3) is
   (a) 25 (b) 5 (c) 3 (d) 15

Answers: 1. c 2. a 3. b 4. a 5. b